Álgebra en FEC (II)

Ferraris Domingo

July 31, 2026

Resumen

Ejercicio 1 Polinomios sobre GF(2)

a) $\boxed{\begin{array}{l} a = 0b1101010 = 1 + x + x^3 + x^5 \\ b = 0b01101101 = x + x^2 + x^4 + x^5 + x^7 \end{array}}$

b) $\boxed{a + b = 1 + x^2 + x^3 + x^4 + x^7 = 0b10111001}$

c) $\boxed{a \cdot b = x + x^3 + x^5 + x^7 + x^9 + x^{12} = 0b0101010101001}$

d) $\boxed{\frac{a \cdot b}{a} = x^7 + x^5 + x^4 + x^2 + x = b}$

Ejercicio 2 División de polinomios

a) $\boxed{\frac{f}{g} = \frac{x^6 + x^5 + x^3 + x^2 + 1}{x^3 + x + 1} \iff q = x^3 + x^2 + x + 1, \quad r = x^2}$

b) $\boxed{f = x^6 + x^5 + x^3 + x^2 + 1 = q \cdot g + r = (x^3 + x^2 + x + 1) \cdot (x^3 + x + 1) + x^2}$

Ejercicio 3 Polinomios irreducibles y primitivos

a) $\boxed{p = x^4 + x^3 + 1 \implies Raices \notin GF(2)}$

b) $\boxed{\frac{x^4 + x^3 + 1}{x^2 + x + 1} \implies r = x \neq 0 \therefore \text{Irreducible sobre GF(2)}}$

c)

Potencia	Polinomica	Tupla
0	0	$0b0000$
1	1	$0b1000$
α^1	α	$0b0100$
α^2	α^2	$0b0010$
α^3	α^3	$0b0001$
α^4	α^3+1	$0b1001$
α^5	$\alpha^3+\alpha+1$	$0b1101$
α^6	$\alpha^3+\alpha^2+\alpha+1$	$0b1111$
α^7	$\alpha^2+\alpha+1$	$0b1110$
α^8	$\alpha^3+\alpha^2+\alpha$	$0b0111$
α^9	α^2+1	$0b1010$
α^{10}	$\alpha^3+\alpha$	$0b0101$
α^{11}	$\alpha^3+\alpha^2+1$	$0b1011$
α^{12}	$\alpha+1$	$0b1100$
α^{13}	$\alpha^2+\alpha$	$0b0110$
α^{14}	$\alpha^3+\alpha^2$	$0b0011$

Ejercicio 4 Suma y producto de polinomios sobre $GF(2^4)$

a) $\boxed{f+g=(\alpha^3x^2+\alpha^7x+\alpha^2)+(\alpha^5x+\alpha^{10})=\alpha^3x^2+\alpha^{14}x+\alpha^8}$

b) $\boxed{f\cdot g=(\alpha^3x^2+\alpha^7x+\alpha^2)\cdot(\alpha^5x+\alpha^{10})=\alpha^8x^3+\alpha^9x^2+\alpha^{12}x+\alpha^{12}}$

Ejercicio 5 División y potencia de polinomios sobre $GF(2^4)$

a) $\boxed{\frac{f}{g}=\frac{\alpha^3x^3+\alpha^9x^2+x+\alpha^6}{x+\alpha^5}\iff q=\alpha^3x^2+\alpha^5x+\alpha^5,\quad r=\alpha^9}$

b) $\boxed{h^2=(\alpha^3x^2+\alpha^{11}x+\alpha^6)^2=\alpha^6x^4+\alpha^7x^2+\alpha^{12}}$

Ejercicio 6 Búsqueda de raíces

a) $\boxed{f=x^2+\alpha^5x+\alpha^9=(x+\alpha^{11})\cdot(x+\alpha^{13})}$

Ejercicio 7 Ortogonalidad sobre GF(2)

a) $\boxed{u\cdot v=(1,1,0,1)\cdot(0,1,1,1)=0\implies \text{u ortogonal a v en GF(2)}}$

b) $\boxed{w=(1,1)\in V_2 \text{ w ortogonal a si mismo sobre GF(2)}}$

Ejercicio 8 Matriz generadora, forma sistemática y matriz de verificación

a) $$\boxed{G=\begin{bmatrix}1&1&0&0&1&1\\0&1&1&1&0&1\\0&0&1&0&1&1\end{bmatrix}\implies G_{sis}=[I_3\mid P]=\begin{bmatrix}1&0&0&1&0&1\\0&1&0&1&1&0\\0&0&1&0&1&1\end{bmatrix}}$$

b) $\boxed{H = [P^T \mid I_3] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}}$

c) $\boxed{G_{sis} H^T = [0]}$

d) $\boxed{u = (1,0,1) \to v = (1,0,1,1,1,0)}$

e) $\boxed{\text{Codeword de la suma igual a la suma de codewords}}$

Desarrollo

Ejercicio 1 Polinomios sobre GF(2)

a) $\boxed{\begin{array}{l} a = 0b1101010 = 1 + x + x^3 + x^5 \\ b = 0b01101101 = x + x^2 + x^4 + x^5 + x^7 \end{array}}$

b) $\boxed{a + b = 1 + x^2 + x^3 + x^4 + x^7 = 0b10111001}$

$$\begin{aligned} a + b &= (1 + x + x^3 + x^5) + (x + x^2 + x^4 + x^5 + x^7) \\ &= 1 + x + x^3 + x^5 + x + x^2 + x^4 + x^5 + x^7 \\ &= 1 + (x + x) + x^2 + x^3 + x^4 + (x^5 + x^5) + x^7 \\ &= 1 + x^2 + x^3 + x^4 + x^7 \\ &= 0b10111001 \end{aligned}$$

c) $\boxed{a \cdot b = x + x^3 + x^5 + x^7 + x^9 + x^{12} = 0b0101010101001}$

$$\begin{aligned} a \cdot b &= (1 + x + x^3 + x^5) \cdot (x + x^2 + x^4 + x^5 + x^7) \\ &= 1 \cdot (x + x^2 + x^4 + x^5 + x^7) + x \cdot (x + x^2 + x^4 + x^5 + x^7) \\ &+ x^3 \cdot (x + x^2 + x^4 + x^5 + x^7) + x^5 \cdot (x + x^2 + x^4 + x^5 + x^7) \\ \\ &= (x + x^2 + x^4 + x^5 + x^7) + (x^2 + x^3 + x^5 + x^6 + x^8) \\ &+ (x^4 + x^5 + x^7 + x^8 + x^{10}) + (x^6 + x^7 + x^9 + x^{10} + x^{12}) \\ \\ &= x + 0 + x^3 + 0 + x^5 + 0 + x^7 + 0 + x^9 + 0 + x^{12} \\ &= x + x^3 + x^5 + x^7 + x^9 + x^{12} \\ &= 0b0101010101001 \end{aligned}$$

d) $\boxed{\frac{a \cdot b}{a} = x^7 + x^5 + x^4 + x^2 + x = b}$

$$
\begin{array}{r|l}
 & x^7 + x^5 + x^4 + x^2 + x \\ \hline
x^5 + x^3 + x + 1 & x^{12} + x^9 + x^7 + x^5 + x^3 + x \\
 & x^{12} + x^{10} + x^8 + x^7 \\ \hline
 & x^{10} + x^9 + x^8 + x^5 + x^3 + x \\
 & x^{10} + x^8 + x^6 + x^5 \\ \hline
 & x^9 + x^6 + x^3 + x \\
 & x^9 + x^7 + x^5 + x^4 \\ \hline
 & x^7 + x^6 + x^5 + x^4 + x^3 + x \\
 & x^7 + x^5 + x^3 + x^2 \\ \hline
 & x^6 + x^4 + x^2 + x \\
 & x^6 + x^4 + x^2 + x \\ \hline
 & 0
\end{array}
$$

Ejercicio 2 División de polinomios

a) $\boxed{\dfrac{f}{g} = \dfrac{x^6 + x^5 + x^3 + x^2 + 1}{x^3 + x + 1} \iff q = x^3 + x^2 + x + 1, \quad r = x^2}$

$$
\begin{array}{r|l}
 & x^3 + x^2 + x + 1 \\ \hline
x^3 + x + 1 & x^6 + x^5 + x^3 + x^2 + 1 \\
 & x^6 + x^4 + x^3 \\ \hline
 & x^5 + x^4 + x^2 + 1 \\
 & x^5 + x^3 + x^2 \\ \hline
 & x^4 + x^3 + 1 \\
 & x^4 + x^2 + x \\ \hline
 & x^3 + x^2 + x + 1 \\
 & x^3 + x + 1 \\ \hline
 & x^2
\end{array}
$$

b) $\boxed{f = x^6 + x^5 + x^3 + x^2 + 1 = q \cdot g + r = (x^3 + x^2 + x + 1) \cdot (x^3 + x + 1) + x^2}$

$$\begin{aligned}
x^6 + x^5 + x^3 + x^2 + 1 &\stackrel{?}{=} (x^3 + x^2 + x + 1) \cdot (x^3 + x + 1) + x^2 \\
&\stackrel{?}{=} x^3 \cdot (x^3 + x + 1) + x^2 \cdot (x^3 + x + 1) \\
&+ x \cdot (x^3 + x + 1) + 1 \cdot (x^3 + x + 1) + x^2 \\
&= x^6 + x^5 + x^3 + x^2 + 1
\end{aligned}$$

Ejercicio 3 Polinomios irreducibles y primitivos

a) $\boxed{p = x^4 + x^3 + 1 \implies Raices \notin GF(2)}$

$$\begin{aligned}
p(0) &= 0^4 + 0^3 + 1 = 1 \neq 0 \\
p(1) &= 1^4 + 1^3 + 1 = 1 \neq 0
\end{aligned}$$

b) $\boxed{\dfrac{x^4 + x^3 + 1}{x^2 + x + 1} \implies r = x \neq 0 \therefore \text{Irreducible sobre GF(2)}}$

$$\begin{array}{r|c}
 & x^2 + 1 \\ \hline
x^2 + x + 1 & x^4 + x^3 + 1 \\
 & x^4 + x^3 + x^2 \\ \cline{2-2}
 & x^2 + 1 \\
 & x^2 + x + 1 \\ \cline{2-2}
 & x \\
\end{array}$$

c)

Potencia	Polinomica	Tupla
0	0	$0b0000$
1	1	$0b1000$
α^1	α	$0b0100$
α^2	α^2	$0b0010$
α^3	α^3	$0b0001$
α^4	$\alpha^3 + 1$	$0b1001$
α^5	$\alpha^3 + \alpha + 1$	$0b1101$
α^6	$\alpha^3 + \alpha^2 + \alpha + 1$	$0b1111$
α^7	$\alpha^2 + \alpha + 1$	$0b1110$
α^8	$\alpha^3 + \alpha^2 + \alpha$	$0b0111$
α^9	$\alpha^2 + 1$	$0b1010$
α^{10}	$\alpha^3 + \alpha$	$0b0101$
α^{11}	$\alpha^3 + \alpha^2 + 1$	$0b1011$
α^{12}	$\alpha + 1$	$0b1100$
α^{13}	$\alpha^2 + \alpha$	$0b0110$
α^{14}	$\alpha^3 + \alpha^2$	$0b0011$

$$
\begin{aligned}
\alpha^4 + \alpha^3 + 1 = 0 \implies \alpha^4 = \alpha^3 + 1 \\
\alpha^0 &= 1 \\
\alpha^1 &= \alpha \\
\alpha^2 &= \alpha^2 \\
\alpha^3 &= \alpha^3 \\
\alpha^4 &= \alpha^3 + 1 \\
\alpha^5 &= \alpha\alpha^4 = \alpha(\alpha^3 + 1) = \alpha^4 + \alpha = (\alpha^3 + 1) + \alpha = \alpha^3 + \alpha + 1 \\
\alpha^6 &= \alpha(\alpha^3 + \alpha + 1) = \alpha^4 + \alpha^2 + \alpha = (\alpha^3 + 1) + \alpha^2 + \alpha = \alpha^3 + \alpha^2 + \alpha + 1 \\
\alpha^7 &= \alpha(\alpha^3 + \alpha^2 + \alpha + 1) = \alpha^4 + \alpha^3 + \alpha^2 + \alpha = (\alpha^3 + 1) + \alpha^3 + \alpha^2 + \alpha = \alpha^2 + \alpha + 1 \\
\alpha^8 &= \alpha(\alpha^2 + \alpha + 1) = \alpha^3 + \alpha^2 + \alpha \\
\alpha^9 &= \alpha(\alpha^3 + \alpha^2 + \alpha) = \alpha^4 + \alpha^3 + \alpha^2 = (\alpha^3 + 1) + \alpha^3 + \alpha^2 = \alpha^2 + 1 \\
\alpha^{10} &= \alpha(\alpha^2 + 1) = \alpha^3 + \alpha \\
\alpha^{11} &= \alpha(\alpha^3 + \alpha) = \alpha^4 + \alpha^2 = (\alpha^3 + 1) + \alpha^2 = \alpha^3 + \alpha^2 + 1 \\
\alpha^{12} &= \alpha(\alpha^3 + \alpha^2 + 1) = \alpha^4 + \alpha^3 + \alpha = (\alpha^3 + 1) + \alpha^3 + \alpha = \alpha + 1 \\
\alpha^{13} &= \alpha(\alpha + 1) = \alpha^2 + \alpha \\
\alpha^{14} &= \alpha(\alpha^2 + \alpha) = \alpha^3 + \alpha^2 \\
\alpha^{15} &= \alpha(\alpha^3 + \alpha^2) = \alpha^4 + \alpha^3 = (\alpha^3 + 1) + \alpha^3 = 1
\end{aligned}
$$

Ejercicio 4 Suma y producto de polinomios sobre $GF(2^4)$

a) $\boxed{f + g = (\alpha^3 x^2 + \alpha^7 x + \alpha^2) + (\alpha^5 x + \alpha^{10}) = \alpha^3 x^2 + \alpha^{14} x + \alpha^8}$

$$
\begin{aligned}
f + g &= (\alpha^3 x^2 + \alpha^7 x + \alpha^2) + (\alpha^5 x + \alpha^{10}) \\
&= \alpha^3 x^2 + (\alpha^7 + \alpha^5)x + (\alpha^2 + \alpha^{10}) \\
&= \alpha^3 x^2 + \big((\alpha^2 + \alpha + 1) + (\alpha^3 + \alpha + 1)\big)x + \big(\alpha^2 + (\alpha^3 + \alpha)\big) \\
&= \alpha^3 x^2 + (\alpha^3 + \alpha^2)x + (\alpha^3 + \alpha^2 + \alpha) \\
&= \alpha^3 x^2 + \alpha^{14} x + \alpha^8
\end{aligned}
$$

b) $\boxed{f \cdot g = (\alpha^3 x^2 + \alpha^7 x + \alpha^2) \cdot (\alpha^5 x + \alpha^{10}) = \alpha^8 x^3 + \alpha^9 x^2 + \alpha^{12} x + \alpha^{12}}$

$$
\begin{aligned}
f \cdot g &= (a^3 x^2 + a^7 x + \alpha^2) \cdot (\alpha^5 x + \alpha^{10}) \\
&= \alpha^8 x^3 + \alpha^{13} x^2 + \alpha^{12} x^2 + \alpha^{17} x + \alpha^7 x + \alpha^{12} \\
&= \alpha^8 x^3 + (\alpha^{13} + \alpha^{12})x^2 + (\alpha^{17} + \alpha^7)x + \alpha^{12} \\
&= \alpha^8 x^3 + (\alpha^{13} + \alpha^{12})x^2 + (\alpha^2 + \alpha^7)x + \alpha^{12} \\
&= \alpha^8 x^3 + (\alpha^2 + 1)x^2 + (\alpha^2 + \alpha^2 + \alpha + 1)x + \alpha^{12} \\
&= \alpha^8 x^3 + \alpha^9 x^2 + \alpha^{12} x + \alpha^{12}
\end{aligned}
$$

Ejercicio 5 División y potencia de polinomios sobre $GF(2^4)$

a) $\boxed{\dfrac{f}{g} = \dfrac{\alpha^3x^3+\alpha^9x^2+x+\alpha^6}{x+\alpha^5} \iff q = \alpha^3x^2+\alpha^5x+\alpha^5, \quad r = \alpha^9}$

$$
\begin{array}{r|l}
 & \alpha^3x^2+\alpha^5x+\alpha^5 \\
\hline
 & \alpha^3x^3+\alpha^9x^2+x+\alpha^6 \\
x+\alpha^5 & \\
 & \alpha^3x^3+\alpha^8x^2 \\
\cline{2-2}
 & \alpha^5x^2+x+\alpha^6 \\
 & \alpha^5x^2+\alpha^{10}x \\
\cline{2-2}
 & \alpha^5x+\alpha^6 \\
 & \alpha^5x+\alpha^{10} \\
\cline{2-2}
 & \alpha^9
\end{array}
$$

b) $\boxed{h^2 = (\alpha^3x^2+\alpha^{11}x+\alpha^6)^2 = \alpha^6x^4+\alpha^7x^2+\alpha^{12}}$

$$
\begin{aligned}
h^2 &= (\alpha^3x^2+\alpha^{11}x+\alpha^6)\cdot(\alpha^3x^2+\alpha^{11}x+\alpha^6) \\
&= \alpha^3x^2(\alpha^3x^2+\alpha^{11}x+\alpha^6)+\alpha^{11}x(\alpha^3x^2+\alpha^{11}x+\alpha^6)+\alpha^6(\alpha^3x^2+\alpha^{11}x+\alpha^6) \\
&= \alpha^6x^4+\alpha^{14}x^3+\alpha^9x^2+\alpha^{14}x^3+\alpha^{22}x^2+\alpha^{17}x+\alpha^9x^2+\alpha^{17}x+\alpha^{12} \\
&= \alpha^6x^4+(\alpha^{14}+\alpha^{14})x^3+(\alpha^9+\alpha^{22}+\alpha^9)x^2+(\alpha^{17}+\alpha^{17})x+\alpha^{12} \\
&= \alpha^6x^4+\alpha^{22}x^2+\alpha^{12} \\
&= \alpha^6x^4+\alpha^7x^2+\alpha^{12}
\end{aligned}
$$

c) Verificar el resultado anterior: elevar al cuadrado cada coeficiente de h(X) por separado y ubicarlo en el doble del grado correspondiente (identidad $(f(X))^2 = f(X^2)$)

$$
\begin{aligned}
h^2 &= (\alpha^3x^2+\alpha^{11}x+\alpha^6)^2 \\
&= (\alpha^3x^2)^2+(\alpha^{11}x)^2+(\alpha^6)^2 \\
&= \alpha^6x^4+\alpha^{22}x^2+\alpha^{12} \\
&= \alpha^6x^4+\alpha^7x^2+\alpha^{12}
\end{aligned}
$$

Ejercicio 6 Búsqueda de raíces

a) $\boxed{f = x^2 + \alpha^5 x + \alpha^9 = (x + \alpha^{11}) \cdot (x + \alpha^{13})}$

$$\begin{aligned} f(\alpha^{11}) &= (\alpha^{11})^2 + \alpha^5(\alpha^{11}) + \alpha^9 \\ &= \alpha^{22} + \alpha^{16} + \alpha^9 \\ &= \alpha^7 + \alpha + \alpha^9 \\ &= (\alpha^2 + \alpha + 1) + \alpha + (\alpha^2 + 1) \\ &= \alpha^2 + \alpha + \alpha + 1 + \alpha^2 + 1 \\ &= 0 \end{aligned}$$

$$\begin{aligned} f(\alpha^{13}) &= (\alpha^{13})^2 + \alpha^5(\alpha^{13}) + \alpha^9 \\ &= \alpha^{26} + \alpha^{18} + \alpha^9 \\ &= \alpha^{11} + \alpha^3 + \alpha^9 \\ &= (\alpha^3 + \alpha^2 + 1) + \alpha^3 + (\alpha^2 + 1) \\ &= \alpha^3 + \alpha^2 + 1 + \alpha^3 + \alpha^2 + 1 \\ &= 0 \end{aligned}$$

Ejercicio 7 Ortogonalidad sobre GF(2)

a) $\boxed{u \cdot v = (1,1,0,1) \cdot (0,1,1,1) = 0 \implies \text{u ortogonal a v en GF(2)}}$

$$\begin{aligned} u \cdot v &= \sum_{i=1}^{4} u_i v_i \pmod 2 \\ &= u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4 \\ &= (1)(0) + (1)(1) + (0)(1) + (1)(1) \\ &= 0 + 1 + 0 + 1 \\ &= 0 \end{aligned}$$

b) $\boxed{w = (1,1) \in V_2 \text{ w ortogonal a si mismo sobre GF(2)}}$

$$\begin{aligned} w \cdot w &= w_1w_1 + w_2w_2 \pmod 2 \\ &= w_1^2 + w_2^2 \end{aligned}$$

Cualquier vector con numero par de 1's cumpliria.

Para este caso, el unico: (1,1)

$$\begin{aligned} &= 1^2 + 1^2 \\ &= 0 \end{aligned}$$

Esto sobre los numeros reales no es posible ya que su producto interno es definido positivo.

Sobre los reales unicamente el elemento nulo es ortogonal a si mismo.

Ejercicio 8 Matriz generadora, forma sistemática y matriz de verificación

a) $\boxed{G = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} \implies G_{sis} = [I_3 \mid P] = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}}$

$G = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$

Comenzamos a operar con fila 2:

$F_2 + F_3 = [0\ 1\ 0\ 1\ 1\ 0] \to F_2$

$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$

Operar con fila 1:

$F_1 + F_3 = [1\ 0\ 0\ 1\ 0\ 1] \to F_1$

$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} = G_{sis}$

b) $\boxed{H = [P^T \mid I_3] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}}$

c) $\boxed{G_{sis}H^T = [0]}$

Tenemos a: $G_{sis} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}, H^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Multiplicamos matrices usando el metodo por sumatoria de filas:

g_1=[1 0 0 1 0 1]

Columnas distintas de cero (1,4,6), entonces:

$$\begin{aligned} &= H_1^T + H_4^T + H_6^T \\ &= (1,0,1) + (1,0,0) + (0,0,1) \\ &= (0,0,0) \end{aligned}$$

g_2=[0 1 0 1 1 0]

Columnas distintas de cero (2,4,5), entonces:

$$\begin{aligned} &= H_2^T + H_4^T + H_5^T \\ &= (1,1,0) + (1,0,0) + (0,1,0) \\ &= (0,0,0) \end{aligned}$$

g_3=[0 0 1 0 1 1]

Columnas distintas de cero (3,5,6), entonces:

$$\begin{aligned} &= H_3^T + H_5^T + H_6^T \\ &= (0,1,1) + (0,1,0) + (0,0,1) \\ &= (0,0,0) \end{aligned}$$

Uniendo los resultados:

$$G_{sis}H^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = [0]$$

d) $\boxed{u = (1,0,1) \rightarrow v = (1,0,1,1,1,0)}$

$$G_{sis} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Buscamos la codeword como:

$v = uG_{sis}$

Usando el metodo por sumatoria de filas:

u=(1,0,1)

Columnas distintas de cero (1,3), entonces:

$$\begin{aligned} v &= G_1 + G_3 \\ &= (1,0,0,1,0,1) + (0,0,1,0,1,1) \\ &= (1,0,1,1,1,0) \end{aligned}$$

Finalmente verificamos que $vH^T = 0$, usando:

$$H^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Usando el metodo por sumatoria de filas:

v=(1,0,1,1,1,0)

Columnas distintas de cero (1,3,4,5), entonces:

$$\begin{aligned} vH^T &= H_1^T + H_3^T + H_4^T + H_5^T \\ &= (1,0,1) + (0,1,1) + (1,0,0) + (0,1,0) \\ &= (1+0+1+0, 0+1+0+1, 1+1+0+0) \\ &= (0,0,0) \end{aligned}$$

e) $\boxed{\text{Codeword de la suma igual a la suma de codewords}}$

$$G_{sis} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Con los vectores: u_1=(1,1,0), u_2=(1,0,0):

| **1.**

Sumando las codewords:

$$\begin{aligned} v_1 &= u_1 G_{sis} \\ &= G_1 + G_2 \\ &= (1,0,0,1,0,1) + (0,1,0,1,1,0) \\ &= (1,1,0,0,1,1) \end{aligned}$$

$$\begin{aligned} v_2 &= u_2 G_{sis} \\ &= G_1 \\ &= (1,0,0,1,0,1) \end{aligned}$$

$$\begin{aligned} v_{3.1} &= v_1 + v_2 \\ &= (1,1,0,0,1,1) + (1,0,0,1,0,1) \\ &= (0,1,0,1,1,0) \end{aligned}$$

| **2.**

En cambio, sumando los vectores u:

$$\begin{aligned} u_3 &= u_1 + u_2 \\ &= (1,1,0) + (1,0,0) \\ &= (0,1,0) \end{aligned}$$

$$\begin{aligned} v_{3.2} &= (u_3) G_{sis} \\ &= G_2 \\ &= (0,1,0,1,1,0) = v_{3.1} \end{aligned}$$

Obtenemos el mismo resultado, por tanto la suma de codewords es igual a la codeword de la suma.
